Queen Mary
University of London
Science and Engineering

EBU4202: Digital Circuit Design
Switching Algebra Tutorial

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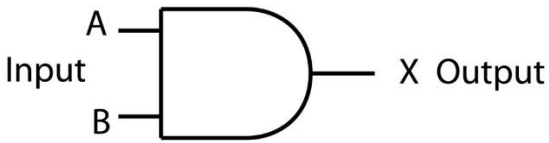
Review Gates

AND



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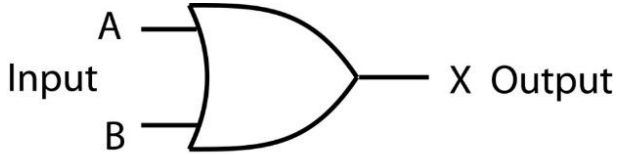

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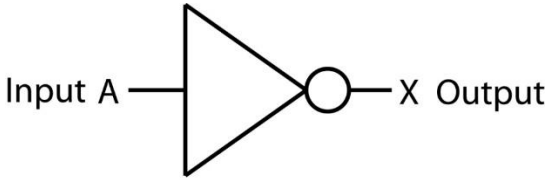
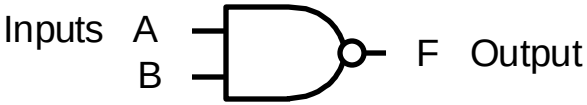


NAND



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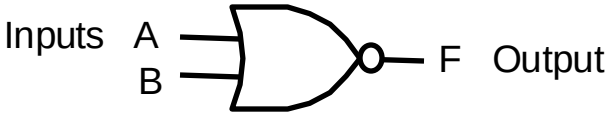
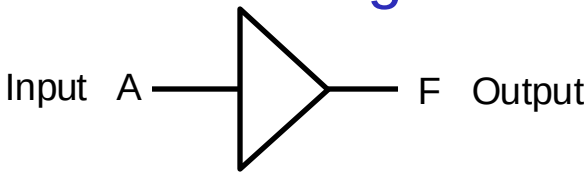
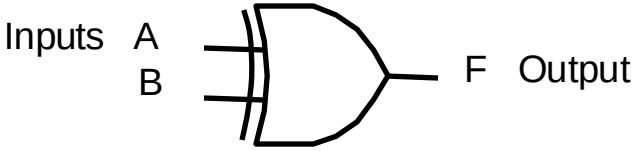


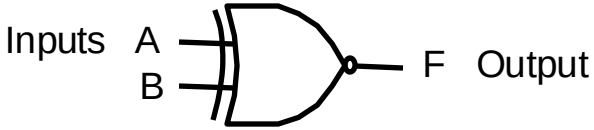
Non-inverting buffer



XOR



XNOR



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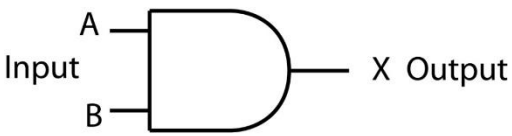
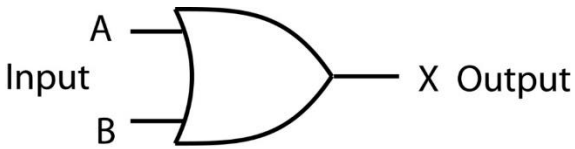


Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

$X = A.B$

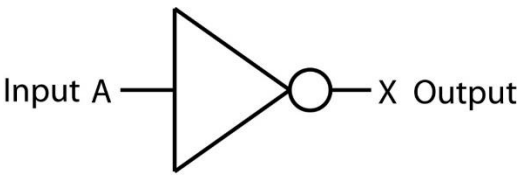
OR



Input		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

$X = A+B$

NOT



Input	Output
A	X
0	1
1	0

$X = A'$

NAND

A	B	F
0	0	1
0	1	1
1	0	1
1	1	0

$F = (A.B)'$

NOR

A	B	F
0	0	1
0	1	0
1	0	0
1	1	0

$F = (A+B)'$

XOR

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

$F = (A \oplus B)$

$F = A'B+AB'$

XNOR

A	B	F
0	0	1
0	1	0
1	0	0
1	1	1

$F = (A \oplus B)'$

$F = A'B'+AB$

Switching Algebra: Basic Definitions

- ***Literal***: variable or the complement of a variable.
  - **Examples**: **X**, **Y**, **X′**, **Y′**
- ***Product Term***: single literal or logical product of 2+ literals.
  - **Examples**: **X**, **X·Y·Z′**, **X·Y′**
- ***Sum of Products Expression (SOP)***: logical sum of product terms
  - **Example**: **X·Z′ + X·Y·Z′ + X·Y′**
- ***Sum Term***: single literal or logical sum of 2+ literals.
  - **Examples**: **X**, **X + Y + Z′**, **X + Y′**
- ***Product of Sums Expression (POS)***: logical product of sum terms
  - **Example**: **(X + Z′)·(X + Y + Z′)·(X + Y′)**

Precedence Rules & Axioms

- **Precedence Rules**: $'$, $\cdot$ and **+**. (invert, and, or)
- **Axioms**:
  - A minimum set of basic mathematic definitions that are assumed to be always **True** and from which we can derive theorems.
  - Let **X** be a logic variable taking on values **0** or **1**.

(A1) $X=0$ if $X\neq 1$	(A1') $X=1$ if $X\neq 0$
(A2) $X=0$ then $X'=1$	(A2') $X=1$ then $X'=0$
(A3) $0\cdot 0=0$	(A3') $1+1=1$
(A4) $1\cdot 1=1$	(A4') $0+0=0$
(A5) $0\cdot 1=1\cdot 0=0$	(A5') $0+1=1+0=1$

Single Variable Theorems & Perfect Induction

- Let **X** be a logic variable:

(T1) X+0 = X	(T1') X·1 = X
(T2) X+1 = 1	(T2') X·0 = 0
(T3) X+X = X	(T3') X·X = X
(T4) (X')' = X	
(T5) X+X' = 1	(T5') X·X' = 0

These theorems can be proved using axioms via ***perfect induction***.

- **Perfect Induction** (better known as ***common sense***).
  - ***Example***:
    - (T1′) X·1 = X

Proof

Hence, X.1=X.

2 possible values for literal ***X***

$$\begin{cases} [X = 0] \rightarrow 0 \cdot 1 = 0 & (\textit{by axiom } \boldsymbol{A5}) \\ [X = 1] \rightarrow 1 \cdot 1 = 1 & (\textit{by axiom } \boldsymbol{A4}) \end{cases}$$

Two (and Three) Variable Theorems

(T6) X + Y = Y + X	**(T6') XY = YX**	***(Commutativity)***
(T7) (X + Y) + Z = X + (Y + Z)	**(T7') (XY)Z = X(YZ)**	***(Associativity)***
(T8) XY + XZ = X (Y + Z)	**(T8') (X + Y)(X + Z) = X + YZ**	***(Distributivity)***
(T9) X + XY = X	**(T9') X(X + Y) = X**	***(Covering)*** Absorption
(T10) XY +XY' = X	**(T10') (X + Y)(X + Y') = X**	***(Combining)***
(T11) XY +X'Z + YZ = XY + X'Z		***(Consensus)***

(T11') (X + Y)(X' + Z)(Y + Z) = (X + Y) (X' + Z)

Adsorption Theorem (not in the textbook):
(T*) X + X'Y = X + Y **(T*)'** X(X' + Y) = XY

Principle of Duality: Any theorem or identity in Switching Algebra remains True if **0's** and **1's** are swapped, and the same for • (**AND**) and **+** (**OR**).

Consensus Theorem (T11)

- Useful in simplifying Boolean expressions
  - Allows the elimination of unnecessary terms.

Truth Table

X	Y	Z	F	G
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	0	0
1	0	1	0	0
1	1	0	1	1
1	1	1	1	1

Show: $XY + \overline{X}Z + YZ = XY + \overline{X}Z$

$$F = XY + \overline{X}Z + YZ = XY + \overline{X}Z + YZ\overbrace{(X+\overline{X})}^{=1}$$

expand $\rightarrow$ $= XY + \overline{X}Z + XYZ + \overline{X}YZ$

reorder $\rightarrow$ $= XY + XYZ + \overline{X}Z + \overline{X}YZ$

factorise $\rightarrow$ $= XY(\cancel{1 + Z}) + \overline{X}Z(\cancel{1 + Y})$

$=1$

$$G = XY + \overline{X}Z$$

$\therefore$ **YZ** is an unnecessary (or redundant) term

Adsorption Theorem

Adsorption Theorem (not in the textbook):
(T*) $X + X'Y = X + Y$ **(T*)'** $X(X' + Y) = XY$

Proof:

$$X + X'Y = (X + X')(X + Y) \quad (T8')$$
$$= 1.(X+Y) \quad (T5)$$
$$= X+Y \quad (T1')$$

(T8') $(X + Y)(X + Z) = X + YZ$

(T5) $X+X' = 1$

(T1') $X \cdot 1 = X$

N Variable Theorems

- **Two or three variables' theorems can be extended** to an arbitrary number of variables, ***N***:
  - ***Generalised Idempotency***

$$(T12)\quad X+X+\ldots+X=X$$
$$(T12')\quad X\cdot X\cdot\ldots\cdot X=X$$

(T14) states: given any *Nvariable* logic expression, its complement can be obtained by swapping **+** and **·**, and complementing all variables.

  - ***De Morgan's Theorems***

$$(T13)\quad (X_1\cdot X_2\cdot\ldots\cdot X_n)'=X_1'+X_2'+\ldots+X_n'$$
$$(T13')\quad (X_1+X_2+\ldots+X_n)'=X_1'\cdot X_2'\cdot\ldots\cdot X_n'$$

(T14): condensed form for **(T13)** & **(T13')**.

  - ***Generalised De Morgan's Theorem***

$$(T14)\quad [F(X_1,X_2,\ldots,X_n,+,\cdot)]'=F(X_1',X_2',\ldots,X_n',\cdot,+)$$

  - ***Shannon's Expansion Theorems***

$$(T15)\quad F(X_1,X_2,\ldots,X_n)=X_1\cdot F(1,X_2,\ldots,X_n)+X_1'\cdot F(0,X_2,\ldots,X_n)$$
$$(T15')\quad F(X_1,X_2,\ldots,X_n)=[X_1+F(0,X_2,\ldots,X_n)]\cdot[X_1'+F(1,X_2,\ldots,X_n)]$$

De Morgan: "Break the line and change the sign"

Idempotency: X+X=X, X.X=X

Example (1/3): Two Equations

- Show that $F_1 = F_2$ using a Truth Table.

$$F_1 = X' \bullet Y' \bullet Z + X' \bullet Y \bullet Z + X \bullet Y'$$

$$F_2 = X' \bullet Z + X \bullet Y'$$

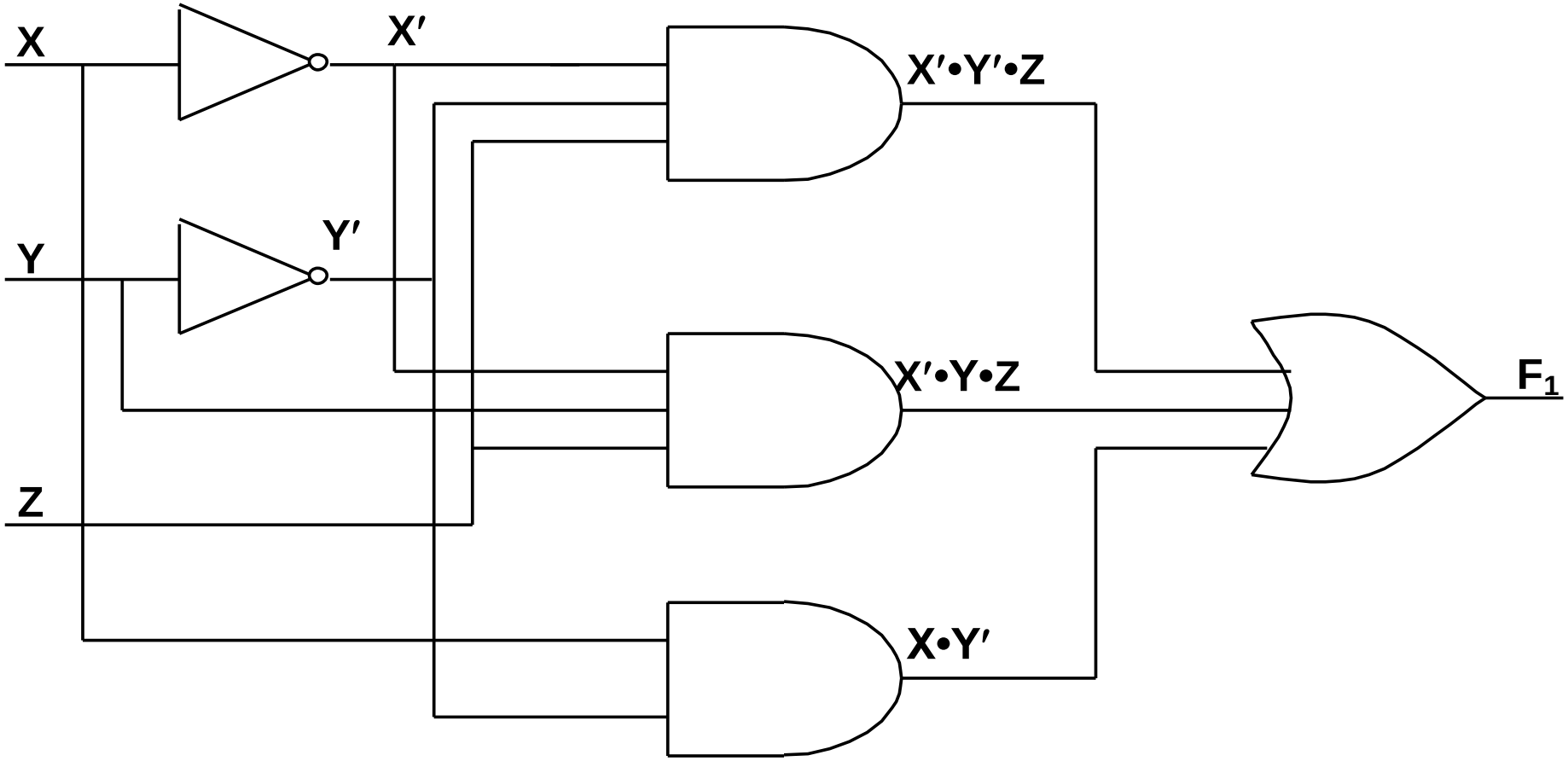
To prove the equality, we can use Switching Algebra theorems **T8** + **T5**.

(T8) → XY + XZ = X(Y + Z)
(T5) → X + X' = 1
Show: prove that $F_1 = F_2$ using Switching Algebra,
F_1 = X'Y'Z + X'YZ + XY' =
= X'Z(Y' + Y) + XY' , using (T8)
= X'Z + XY' , using (T5)
= F_2

X	Y	Z	F_1	F_2
0	0	0	0	0
0	0	1	1	1
0	1	0	0	0
0	1	1	1	1
1	0	0	1	1
1	0	1	1	1
1	1	0	0	0
1	1	1	0	0

Example (2/3): Two Equations

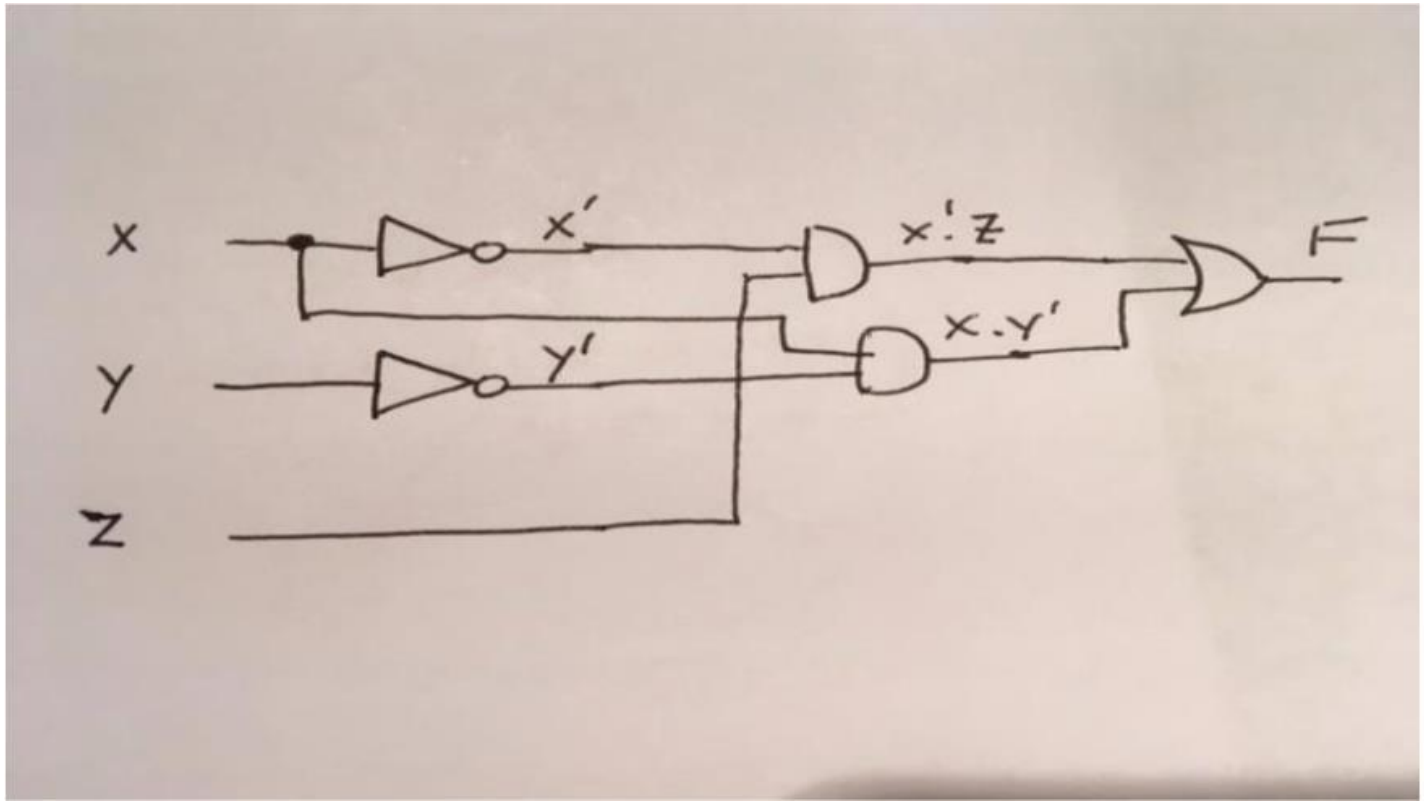
$F_1 = X' \bullet Y' \bullet Z + X' \bullet Y \bullet Z + X \bullet Y'$



Example (3/3): Two Equations

To be completed in class …

$F_2 = X' \bullet Z + X \bullet Y'$



Example 1: Minimisation

F = A′B′C + AB′C

⇕ (T10) – Combining Theorem XY+XY'=X

F = B′C

How is (T10) applied here?

Alternatively, if we were to factor ***A*** out, it would have left us with:

F = B′C(A′ + A) **(using T8)**

A′ + A = 1 **(using T5)**

therefore, **F = B′C**

(Proof of T10)

Example 2: Minimisation (1)

$F = A'B'C' + A'B'C + A'BC$

$A'B'C = A'B'C + A'B'C$

(T3) $X + X = X$

$F = A'B'C' + A'B'C + A'B'C + A'BC$

T10 **T10**

$F = A'B' + A'C$

(T10) $XY + XY' = X$

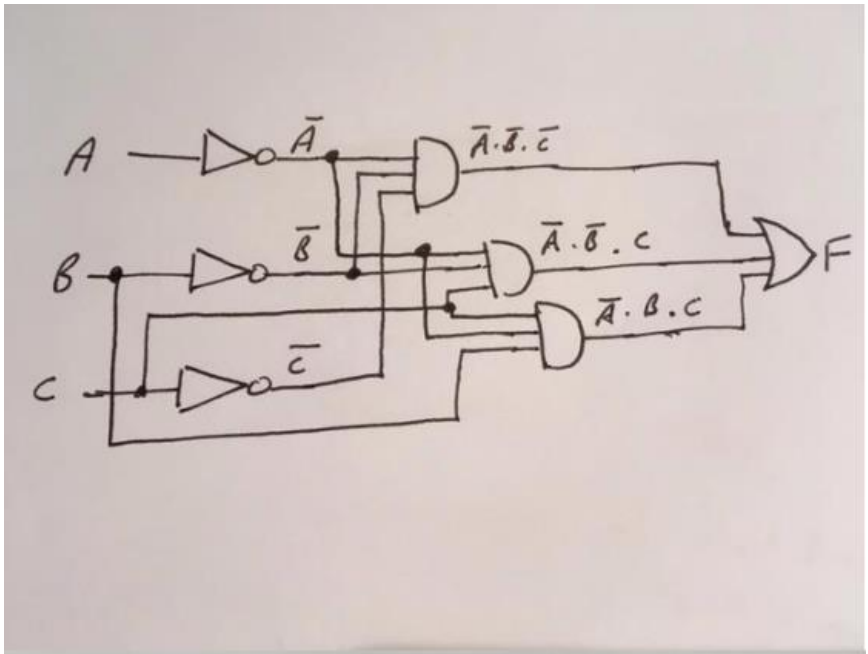
Circuit Diagrams for these functions?

Note: Can be further simplified to F = A'(B'+C)

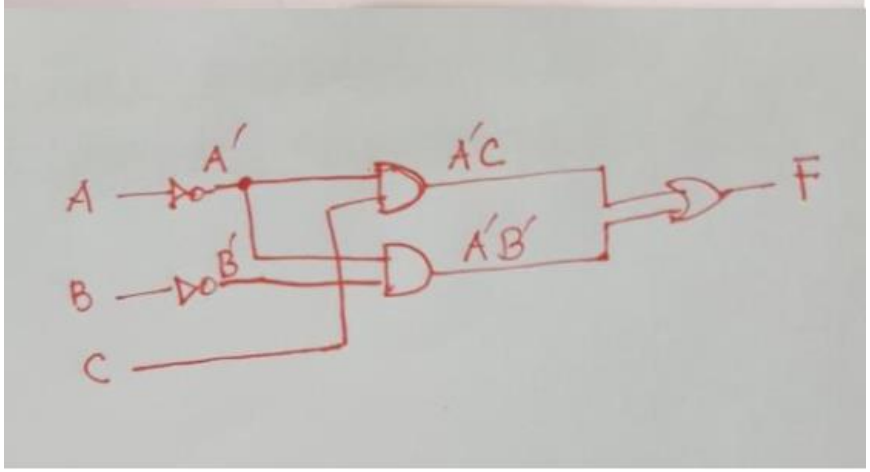
Example 2: Minimisation (2)

To be completed in class …

F = A′B′C′ + A′B′C + A′BC



F = A′B′ + A′C



Example: Minimisation of F

Simplify $F = (AB'(A + C))' + A'B(A + B' + C)'$

$= (AB' + AB'C)' + A'B(A + B' + C)'$

MULTIPLY THROUGH TERM 1 APPLYING (T3′) X•X=X

APPLY (T13′) to $(AB' + AB'C)'$ & $(A + B' + C)'$

$= (A' + B)(A' + B + C') + A'B(A'BC')$

EXPAND TERMS

$= (A' + B)(A' + B + C') + A'BC'$

$= A'A' + A'B + A'C' + BA' + BB + BC' + A'BC'$

APPLY (T3′)

$= A' + A'B + A'C' + BA' + B + BC' + A'BC'$

APPLY (T1′)

$= A'(1 + B + C' + B + BC') + B(1 + C')$

APPLY (T2)

$= A'(1) + B(1)$

$= A' + B$

Minterms & ***Maxterms*** for 3 Variables

X	Y	Z	Product Term	Symbol	Sum Term	Symbol
0	0	0	$\overline{X}\cdot\overline{Y}\cdot\overline{Z}$	m_0	$X+Y+Z$	M_0
0	0	1	$\overline{X}\cdot\overline{Y}\cdot Z$	m_1	$X+Y+\overline{Z}$	M_1
0	1	0	$\overline{X}\cdot Y\cdot\overline{Z}$	m_2	$X+\overline{Y}+Z$	M_2
0	1	1	$\overline{X}\cdot Y\cdot Z$	m_3	$X+\overline{Y}+\overline{Z}$	M_3
1	0	0	$X\cdot\overline{Y}\cdot\overline{Z}$	m_4	$\overline{X}+Y+Z$	M_4
1	0	1	$X\cdot\overline{Y}\cdot Z$	m_5	$\overline{X}+Y+\overline{Z}$	M_5
1	1	0	$X\cdot Y\cdot\overline{Z}$	m_6	$\overline{X}+\overline{Y}+Z$	M_6
1	1	1	$X\cdot Y\cdot Z$	m_7	$\overline{X}+\overline{Y}+\overline{Z}$	M_7

Maxterms

is the sum of ***n*** distinctive logic variables or their complements

Minterms

a product of ***n*** distinctive logic variables (or their complements);

Representing a Function in *Standard Form*

- Logic Functions' Representations:
  - ***Truth Table***: practical only for a small number of variables.
  - ***Algebraic sum of minterms*** (*Canonical Sum*)
  - ***Algebraic product of maxterms*** (*Canonical Product*)

Truth Table

X	Y	Z	F	$\overline{F}$	
0	0	0	1	0	m_0
0	0	1	0	1	m_1
0	1	0	1	0	m_2
0	1	1	0	1	m_3
1	0	0	0	1	m_4
1	0	1	1	0	m_5
1	1	0	0	1	m_6
1	1	1	1	0	m_7

So: $F = X'\bullet Y'\bullet Z' + X'\bullet Y\bullet Z' + X\bullet Y'\bullet Z + X\bullet Y\bullet Z$

$F = \mathbf{m_0} + \mathbf{m_2} + \mathbf{m_5} + \mathbf{m_7} = \Sigma m(0, 2, 5, 7)$

And: $F' = \mathbf{m_1} + \mathbf{m_3} + \mathbf{m_4} + \mathbf{m_6} = \Sigma m(1, 3, 4, 6)$

$F'' = (\mathbf{m_1} + \mathbf{m_3} + \mathbf{m_4} + \mathbf{m_6})'$

$F'' = \mathbf{m_1}' \bullet \mathbf{m_3}' \bullet \mathbf{m_4}' \bullet \mathbf{m_6}'$ de Morgan

$F = \mathbf{M_1} \bullet \mathbf{M_3} \bullet \mathbf{M_4} \bullet \mathbf{M_6} = \prod M(1, 3, 4, 6)$

Example: Sum of Products & Simplification (1/2)

A Boolean Function in ***Sum of Minterms*** form can be simplified to a ***Sum of Products*** form by algebraic manipulation or map simplification.

Truth Table

X	Y	Z	F	
0	0	0	1	m_0
0	0	1	1	m_1
0	1	0	1	m_2
0	1	1	0	m_3
1	0	0	1	m_4
1	0	1	1	m_5
1	1	0	0	m_6
1	1	1	0	m_7

F = 1 for each of the combinations of variables:
X'•Y'•Z', X'•Y'•Z, X'•Y•Z', X•Y'•Z', X•Y'•Z

So: from the truth table:

F = X'•Y'•Z' + X'•Y'•Z + X'•Y•Z' + X•Y'•Z' + X•Y'•Z

F = Σm(0, 1, 2, 4, 5)

∴ ***F*** can be simplified by algebraic or map simplification techniques to a ***Sum of Products***.

Then can use Boolean Algebra to simplify the function…..

Example: Sum of Products & Simplification (2/2)

Which equation is easier to build?

$$F = X'\bullet Y'\bullet Z' + X'\bullet Y'\bullet Z + X'\bullet Y\bullet Z' + X\bullet Y'\bullet Z' + X\bullet Y'\bullet Z$$

T10 T10

$$F = X'\bullet Y' + X'\bullet Y\bullet Z' + X\bullet Y'$$

T10

$$F = Y' + X'\bullet Y\bullet Z'$$

T* (Adsorption Theorem)

$$F = Y' + X'\bullet Z'$$

Simplification of **F** using algebraic manipulation.

Exercise

Simplify the following two functions:

a) F=AB’(C+D)+C’D’

b) G=(A+B)(A+C’)(A+D)(BC’D+E)

Exercise

Solutions……

a)

$F=AB'(C+D)+C'D'$

$= AB'(C'D')'+C'D'$ [de Morgan's Theorem]

$=AB'+C'D'$ [T8']

b)

$G=(A+B)(A+C')(A+D)(BC'D+E)$

$=(A+BC')(A+D)(E+BC'D)$ [T8': $(X+Y)(X+Z)=X+YZ$]

$=(A+BC'D)(E+BC'D)$ [T8': $(X+Y)(X+Z)=X+YZ$]

$=AE+BC'D$ [T8']

Minterm & *Maxterm* Expansions of *F*, *F'*

Example for F(X, Y, Z): 2^3 terms (numbered *0* through to *7*).

Given Form Example	Minterm Expansion of F	Maxterm Expansion of F	Minterm Expansion of F'	Maxterm Expansion of F'
F = ∑m(3,4,5,6,7)	–	∏M(0,1,2)	∑m(0,1,2)	∏M(3,4,5,6,7)
F = ∏M(0,1,2)	∑m(3,4,5,6,7)	–	∑m(0,1,2)	∏M(3,4,5,6,7)

Expansion into Canonical SOP & POS

- How to **expand Boolean function into canonical SOP**:
  - Use properties (T1') and (T5).
  - Take each *product term* with a missing literal variable (e.g., A) and **AND** it (logic operation) with the *sum term* (A + A').

- How to **expand Boolean function into canonical POS**:
  - Use the *distributive property* i.e., (T8').
  - Take each *sum term* with a missing literal variable (e.g., A) and **OR** it (logic operation) with the *product term* (A.A').

Example: Sum of Products

Problem: Expand the boolean function **F(A,B,C) = A′B′ + BC** into a standard *sum of products* (*aka minterm expansion*, *canonical SOP*).

```
F(A,B,C) = A′B′ + BC
         = A′B′(C+C′) + BC(A+A′)   by (T5), (T1')
         = A′B′C + A′B′C′ + ABC + A′BC   by (T8), (T6), (T6')
```

To get the Σm expression, replace all *normal literals with a 1* and all *primed literals with a 0*: (***Example***: A′B′C ⇒ A′ = 0, B′ = 0, C = 1 ⇒ 001).

F(A,B,C) = Σm(001, 000, 111, 011)

Now convert to decimal (and put them in order), F(A,B,C) = Σm(0, 1, 3, 7).

Example: Product of Sums

Problem: Expand the boolean function **F(A,B,C) = A′B′ + BC** into a standard *product of sums* (*aka maxterm expansion*, *canonical POS*).

$$
\begin{aligned}
F(A,B,C) &= A'B' + BC \\
&= (A'B'+B)(A'B'+C) && \text{by variable substitution, (T8')} \\
&= (A'+B)(B'+B)(A'+C)(B'+C) && \text{by (T8')} \\
&= (A'+B+CC')(A'+C+BB')(B'+C+AA') && \text{by (T1), (T5')} \\
&= (A'+B+C)(A'+B+C')(A'+C+B)(A'+C+B')(B'+C+A)(B'+C+A') \\
&= (A'+B+C)(A'+B+C')(A'+B'+C)(A+B'+C) && \text{by (T3'), (T6)}
\end{aligned}
$$

To get the ΠM expression replace all *normal literals with a 0* and all *primed literals with a 1*: (***Example***: A′+B′+C ⇒ A′ = 1, B′ = 1, C = 0 ⇒ 110).

F(A,B,C) = ΠM (100, 101, 110, 010)

Now convert to decimal (and order them): F(A,B,C) = ΠM (2, 4, 5, 6).

Deriving the Standard *Product of Sums* Expression

- ***Remember***: If you are unsure of your answer, you can check it with a Truth Table; fill in Truth Table as per original **F = A'B' + BC**!

For: **F'** (*minterms*) or **F** (*maxterms*) where **F = 0**.

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

For: **F** (*minterms*) or **F'** (*maxterms*) where **F = 1**.

Does our expression have all the terms indicated by the Truth Table?

F = (A'+B+C)(A'+B+C').
(A'+B'+C)(A+B'+C)